AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *[GEN26-A0H-I065]*

Team Name: *[The 4 Dimension]*

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Presentation Outline

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Problem Statement

[To develop an app that detects the pothole, especially during rain which detects the depth of the pothole and provides timely alerts to road users and prevent accidents and improve road safety.]

Problem Understanding

- [Drivers, motorcyclists, cyclists, and pedestrians who travel on pothole-damaged roads, especially during rain.]
- [Heavy rainfall, poor drainage, water accumulation, and road surface damage make potholes difficult to detect and judge.]
- [It causes accidents, vehicle damage, injuries, traffic congestion, and risks to road users due to late detection of potholes.]

Existing Gaps & Challenges

- [Existing pothole detection systems may struggle to identify potholes hidden by rainwater or poor visibility.]
- [Many systems detect potholes but may not accurately estimate their depth or provide timely warnings to road users.]
- [Accurately measuring the depth of water-filled potholes is difficult due to the reflective water surface.]

Our Proposed Solution

- [A smartphone-based pothole detection app that uses accelerometer and gyroscope sensors to detect potholes, estimate their severity, and provide timely alerts to road users, even during rainy conditions.]
- [How the Solution Works 1. Detect: Mobile sensors identify potholes using vehicle movements. 2. Estimate: Assess pothole severity based on sensor data. 3. Locate: GPS marks the pothole location. 4. Alert: Notify road users about potholes, especially during rain.]
- [Main Value Delivered to Users - Early warnings about potholes, especially during rain. - Improved road safety and reduced accidents. - Protection from vehicle damage. - Safer and smoother journeys.]

Technical Approach

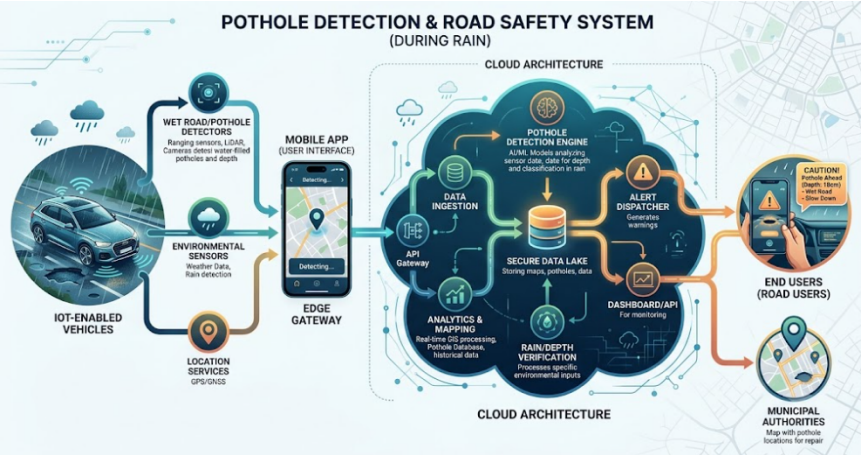


Figure: Pothole detection and road safety system

[Explain how the system components work together.]

Tech Stack

Frontend: [Flutter, React.js]

Database: [PostgreSQL+PostGIS]

Backend: [Python Fast API]

Innovation & Uniqueness

- [What is new or different? Uses smartphone accelerometer and gyroscope sensors without requiring cameras.
  Focuses on detecting potholes even during rainy conditions.
  Estimates pothole severity and tags its GPS location.
  Provides timely alerts to road users to improve safety.]

- [Why is this approach suited to the problem? - Uses readily available smartphone sensors, reducing hardware costs. - Works without image processing, even in low-visibility conditions. - Helps estimate pothole severity and identify its location. - Provides timely alerts to improve road safety during rain.]

- [What is the key differentiator Our key differentiator is camera-free pothole detection using smartphone motion sensors, enabling low-cost detection and timely alerts even during rainy conditions.]

Feasibility & Scalability

- **Feasibility:** [Resources and steps needed to build it: 1. Collect Data: Record sensor readings while travelling on roads. 2. Detect Potholes: Analyse sudden changes in motion data. 3. Estimate Severity: Classify potholes based on sensor patterns. 4. Map Location: Use GPS to record pothole locations. 5. Send Alerts: Notify nearby road users through the app. 6. Test Improve: Validate the app on normal and rainy roads.]

- **Scalability:** [How it can grow or expand? Expand to more cities by partnering with government authorities, integrating real-time pothole maps, and scaling the app to reach more road users.]

- **Risks:** [Main constraints and possible mitigations: Constraints: Sensor inaccuracies, GPS errors, and difficulty estimating water-filled pothole depth. Mitigations: Filter sensor noise, verify GPS locations, and validate severity estimates through real-road testing.]

Cost for Product (Optional)

- Development or hardware cost: [500–7,000]
- Hosting or maintenance cost: [500–3,000/month]
- Estimated total: [1,000–10,000 for initial development and hardware, plus 500–3,000 per month for hosting and maintenance.]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: [Confusion between pothole alerts and normal notifications]
- Use case 2: [Phone placement affecting sensor readings]
- Use case 3: [Battery consumption by GPS.]

Impacts and Benefits

- [Benefit to the target users: Early pothole alerts, improved road safety during rain, reduced vehicle damage, and safer, smoother journeys.]

- [Social, economic, or environmental impact:Social: Reduces accidents and improves road safety.
  Economic: Minimizes vehicle repair and accident-related costs.
  Environmental: Reduces fuel waste and emissions caused by traffic congestion.]

- [Expected measurable outcome: Detect at least 80
  Provide timely alerts with GPS locations.
  Reduce missed pothole warnings.
  Improve detection reliability on rainy roads.]

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG [11]	Target [11.2]	[Sustainable Cities and Communities]	[Improves road safety for drivers, cyclists, and other road users.]
SDG [9]	Target [9.1]	[Industry, Innovation and Infrastructure]	[Supports safer, more reliable road infrastructure using smartphone technology.]
SDG [3]	Target [3.6]	[Good Health and Well-being]	[Helps reduce road accidents and injuries through timely pothole alerts.]

Individual Member Contribution

Member	Contribution or responsibility
[Shobika S]	[Team leader- UI/UX designer]
[Swathi Theertha K B]	[GPS and map developer]
[Ananya Viswanathan]	[Dtatabase and App logic developer]
[V S Lithanya Varshigha]	[Testing and documentation lead]

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